

Formative 2: HMM Activity Recognition

1. Background and Motivation

I built a smartphone activity monitor to detect standing, walking, jumping, and still states from accelerometer and gyroscope data. This supports fitness tracking and sedentary reminders without extra hardware. The true activity is hidden behind noisy sensors, so I used a Hidden Markov Model to infer activity states over time.

2. Data Collection and Preprocessing

I used Sensor Logger to record 10 sessions per activity (standing, walking, jumping, still). Each recording lasted 5-10 seconds. Files are stored in data/ with accel and gyro CSV pairs. My sampling rate is about 99.9 Hz on one phone, so I did not resample. I used 1-second windows with 0.5-second steps (99 samples per window at 100 Hz). walking_07_accel.csv was missing and was excluded from training.

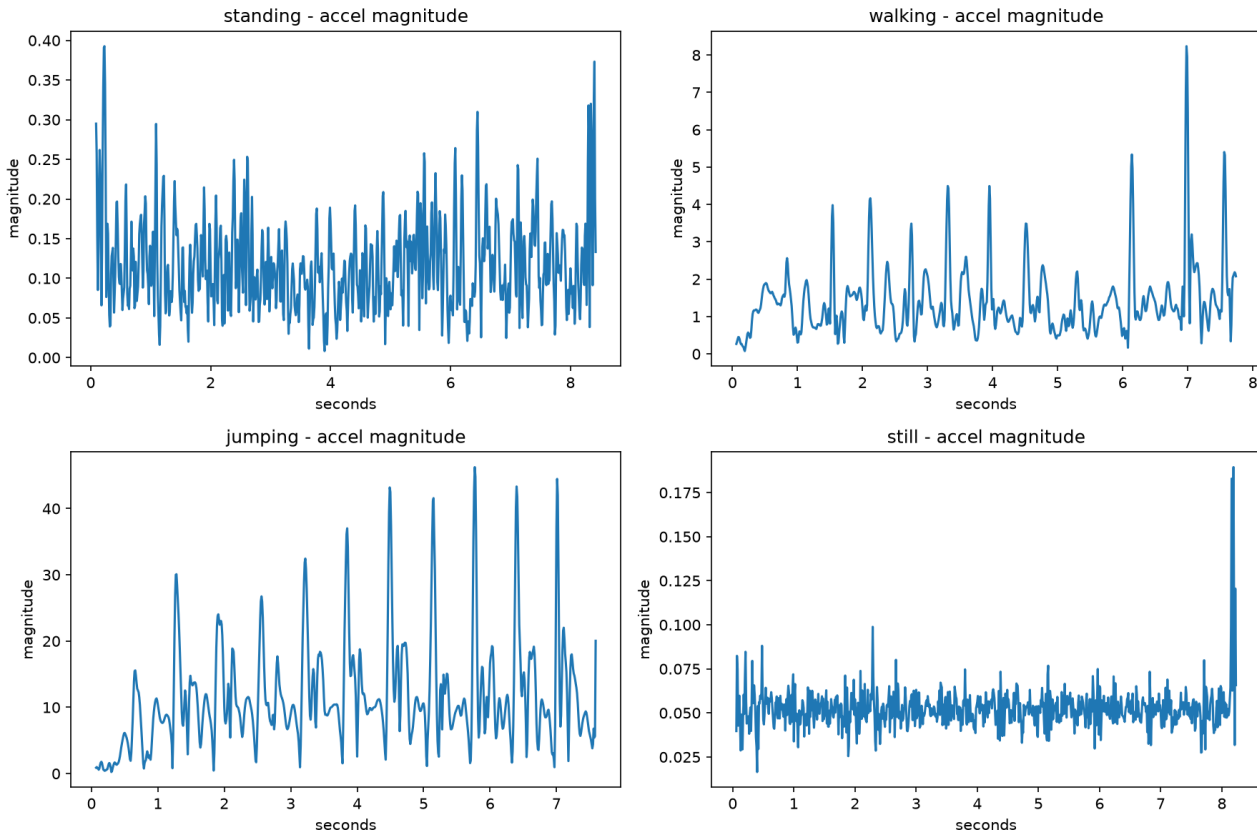


Figure 1: Raw accelerometer magnitude per activity.

3. Feature Extraction

I extracted 8 features per window: mean_mag, std_mag, var_mag, sma, corr_xy (time domain), and dom_freq, spec_energy, fft_peak (FFT on accel magnitude). I applied z-score normalization using training statistics only so large features do not dominate the HMM.

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4. HMM Setup and Implementation

I used a 4-state Gaussian HMM (hmmlearn). Hidden states map to the four activities. Observations are 8D feature vectors. Baum-Welch training used $\text{tol}=1\text{e-}4$ and converged in 2 iterations. Viterbi decoding (model.predict) finds the most likely state sequence. I trained on recordings 01-08 and tested on unseen recordings 09-10.

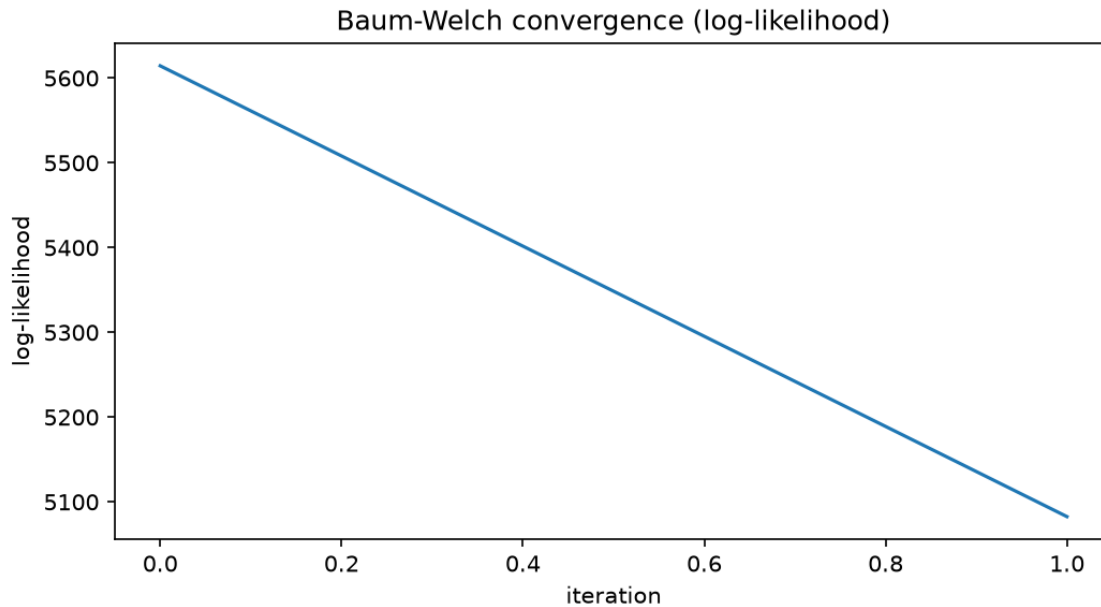


Figure 2: Baum-Welch log-likelihood convergence.

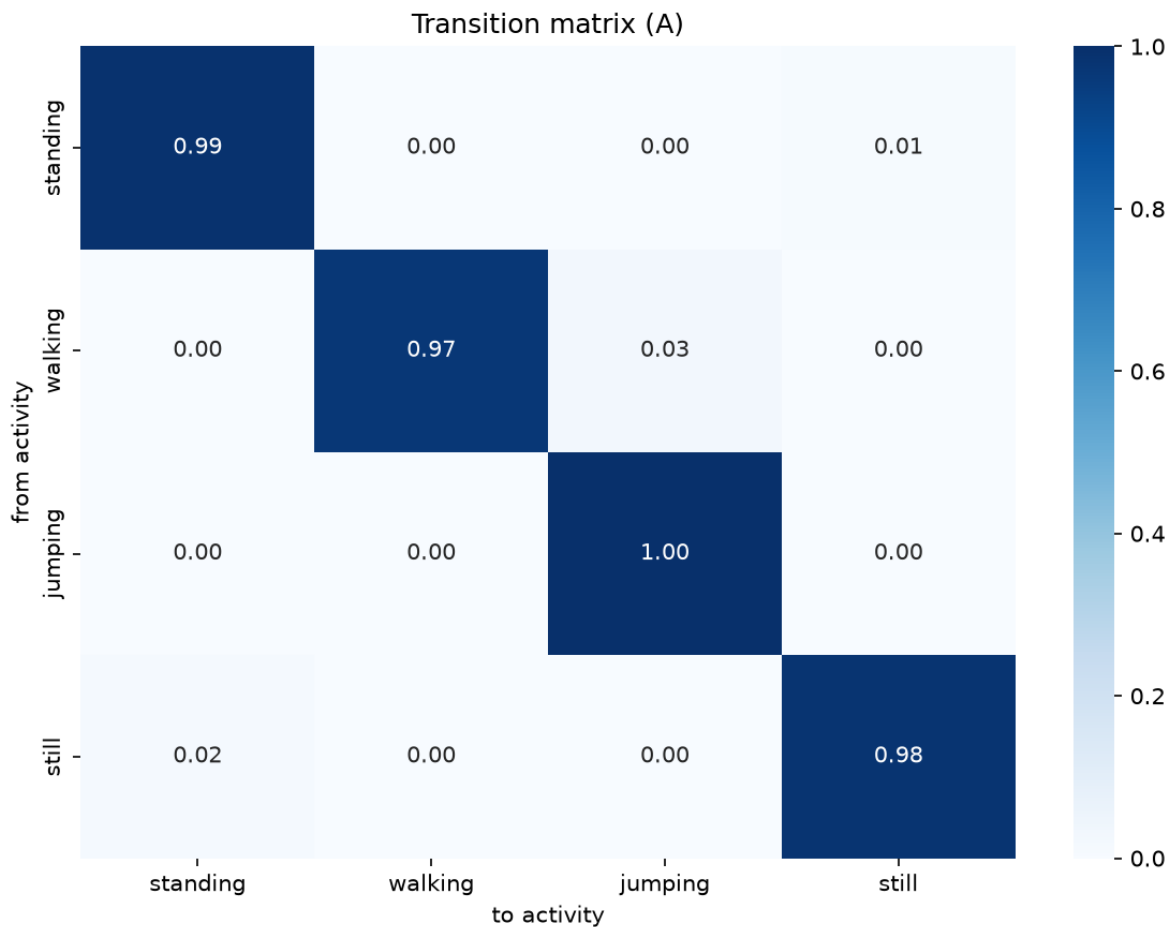


Figure 3: Transition probability matrix (A).

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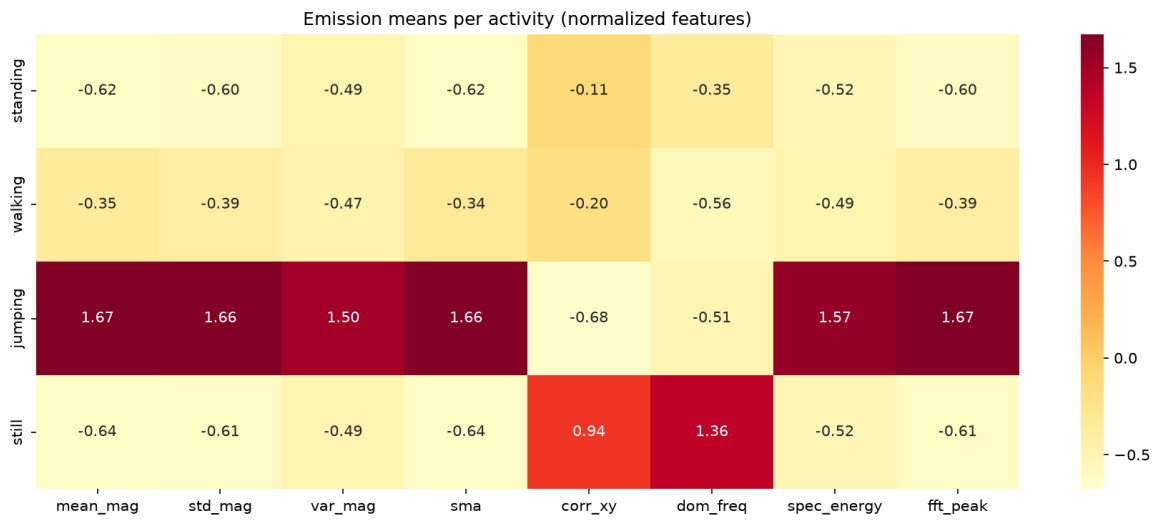


Figure 4: Emission means per activity state (B).

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5. Results and Interpretation

On unseen test data (recordings 09 and 10), the model achieved 99.6% window-level accuracy. Jumping and still were easiest to separate. Standing was hardest and had one misclassified window.

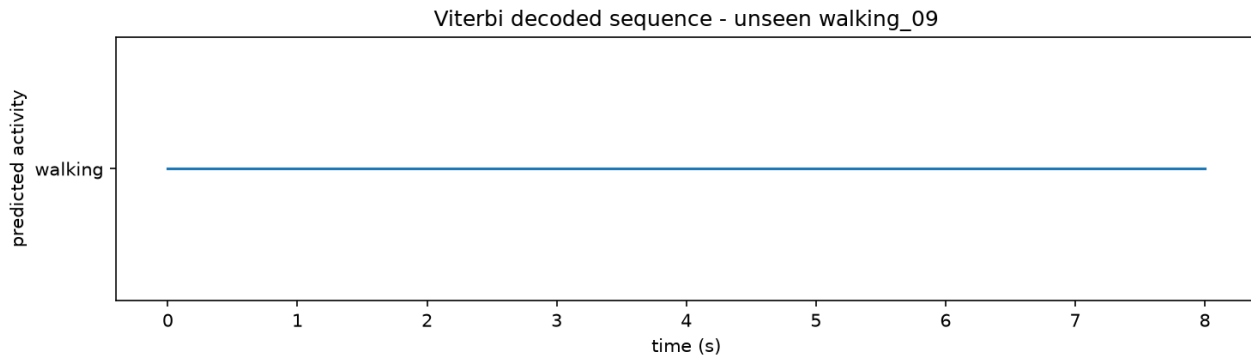


Figure 5: Viterbi decoded sequence on unseen walking_09.

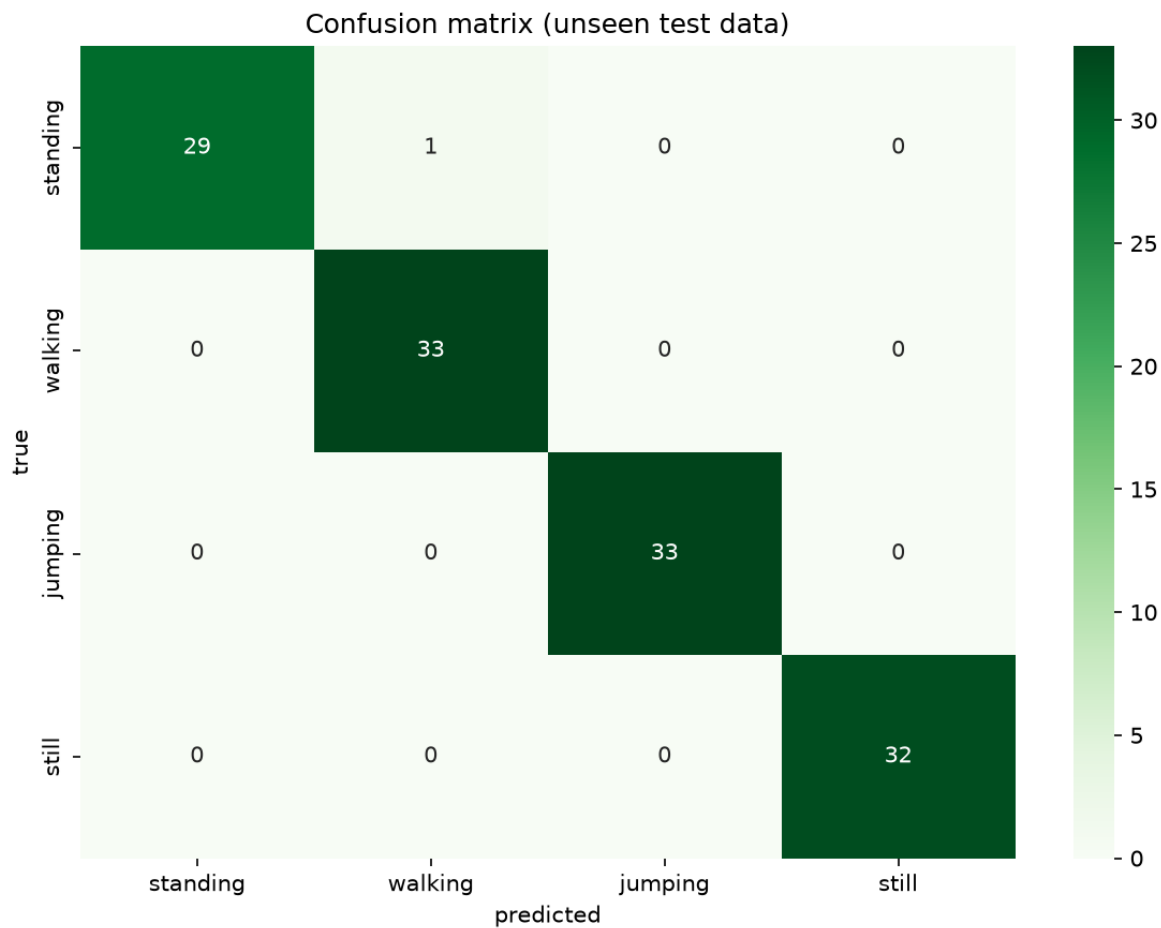


Figure 6: Confusion matrix on unseen test windows.

Evaluation Metrics (Unseen Test Data)

Activity	Samples	Sensitivity	Specificity	Accuracy
standing	30	0.967	1.000	0.992
walking	33	1.000	0.989	0.992
jumping	33	1.000	1.000	1.000
still	32	1.000	1.000	1.000

6. Discussion and Conclusion

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Transition probabilities show high self-loop values, which matches short clips where activity rarely changes. Sensor noise had limited impact because features average over 1-second windows. Improvements: re-record missing walking_07, collect more data, add sensors, and test in new environments.

7. Task Allocation (Individual Work)

I completed all tasks: data collection, cleaning, feature extraction, HMM training, evaluation, visualizations, and this report.